Google Maps Workflow / Flow Chart

From Real-World Data to Your Screen

1 Data Collection

Gathering real-world geographic and traffic data from multiple sources



Street View Cars

Capture 360° images of roads, buildings and surroundings



Satellites

Provide high-resolution aerial and satellite imagery



User Devices

Send real-time location, speed and traffic data (anonymized)

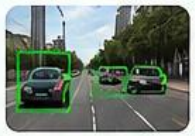


Third-party Sources

Government maps, local business data, public transit data, etc.

2 Image Processing & AI

Convert raw data into useful, structured geographic information



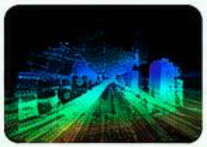
Computer Vision

- Object detection (roads, signs, buildings)
- Image stitching (360°)
- Feature extraction (lanes, landmarks)



Image Analysis

- Detect roads and lanes
- Identify buildings, land use
- Update map features



3D Reconstruction

- Create 3D maps and point clouds
- Generate Street View panoramas



Geotagging

- Assign GPS coordinates
- Align images with map data

3 Data Storage & Processing

Store and manage massive amounts of map and image data



Google Cloud Infrastructure

- Store petabytes of images and map data
- Distributed storage and processing (e.g. BigQuery)
- Continuously update global maps



Map Database

- Roads and lanes
- Buildings and landmarks
- Points of interest (POIs)
- Real-time traffic data
- 3D models and imagery

4 Route Planning & Optimization

Use AI and algorithms to find the best route based on real-time and historical data



Analyze Inputs

- Start and destination
- Real-time traffic conditions
- Road closures, accidents
- User preferences (fastest, shortest, eco-friendly, etc.)

Routing Algorithm

- Uses graph algorithms (e.g. Dijkstra, A*)
- Considers real-time traffic (AI/ML)
- Predicts travel time
- Suggests alternate routes



Optimized Route

- Best route with turn-by-turn directions
- Real-time updates if conditions change

5 Map Rendering

Convert processed data into an interactive map for users



Map Tiles

- Convert map data into image tiles (different zoom levels)
- Store and deliver via global content delivery network (CDN)



Layers

- Satellite view, map view, terrain view
- Traffic layer, transit layer, etc.
- 3D buildings and Street View

6 User Interaction

Deliver maps, routes and real-time updates to users on multiple devices



- **Mobile App** (Android & iOS)
- **Web Browser** (maps.google.com)
- **Android Auto / Apple CarPlay**
- **Wear OS** (Smartwatches)
- **Smart Displays** (e.g. Google Nest)

7 Continuous Feedback & Improvement

Learn from user data and real-world changes to keep maps accurate

User Feedback

- Report wrong locations
- Suggest business edits
- Share traffic incidents

Google Maps

Data Analysis (AI/ML)

- Detect changes in roads, buildings, traffic patterns
- Update maps automatically

Map Updates

- Add new roads and places
- Improve accuracy
- Roll out updates globally